Task result overview

Task information

Software version: 1.2.8

Generated UTC: 2026-10-10T18:10:47Z

Task name: reference_network_report_demo

Analysis duration: 2 h

Time step: 60 s

Thermal network: reference_11_node_gnss_receiver

Task ID: fcdf4fcc-e47f-4b25-b935-d6425021b817

Outputs from this run

scenario, sweep

Main findings

Mean beta angle: 19.46 deg

Eclipse fraction: 28.93 %

RF temperature span: 13.69 K

Semi-synthetic code-bias span: 2.685 m

Peak delay 2201.2 s (1 eligible cycles).

Completed sweep cases: 6 / 6

Maximum energy closure: 2.001e-09 J

Orbit heating and nodal response

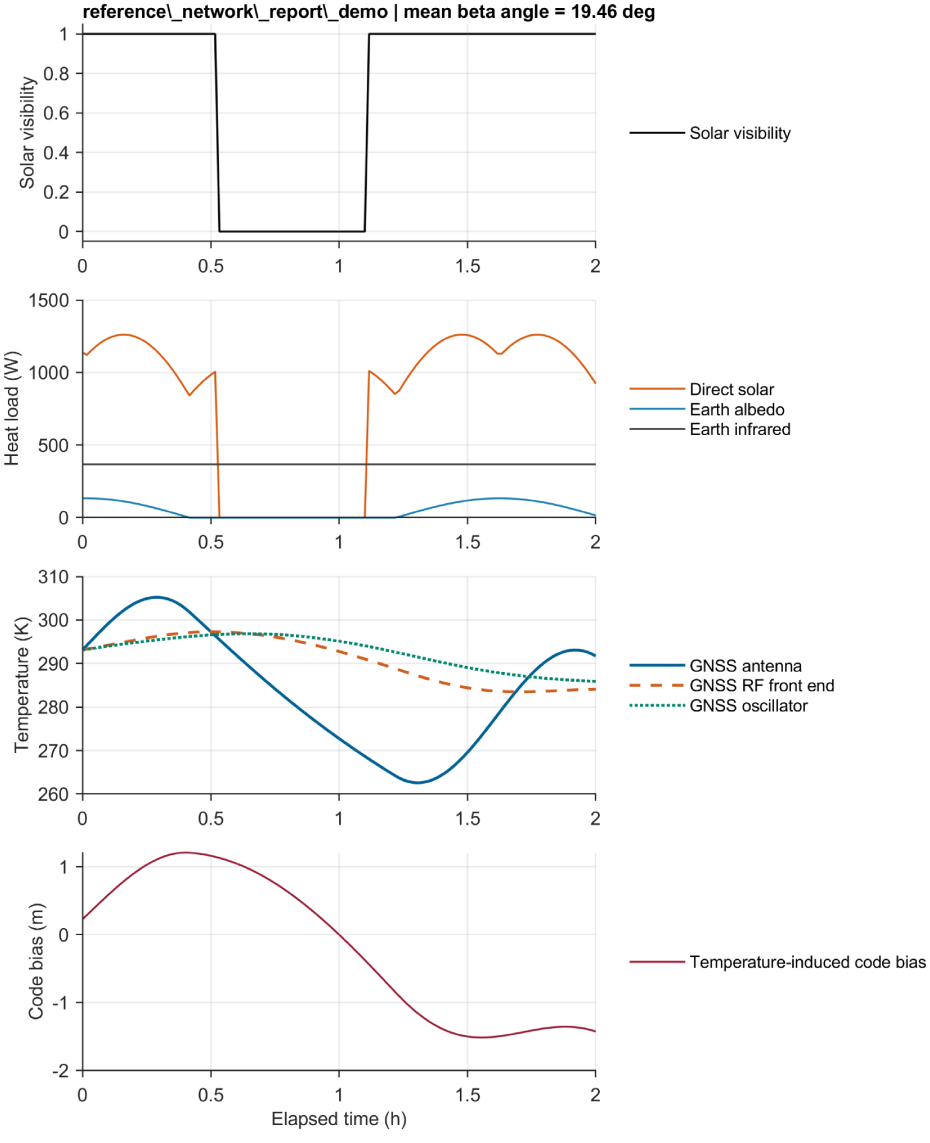
Mean beta angle: 19.46 deg; eclipse fraction: 28.93%.

RF temperature span: 13.69 K; assumed temperature-induced code-bias span: 2.685 m.

Peak delay 2201.2 s (1 eligible cycles).

2 illumination transitions occurred over this run.

No warmup was used; initial-temperature transients may affect spans and lag.



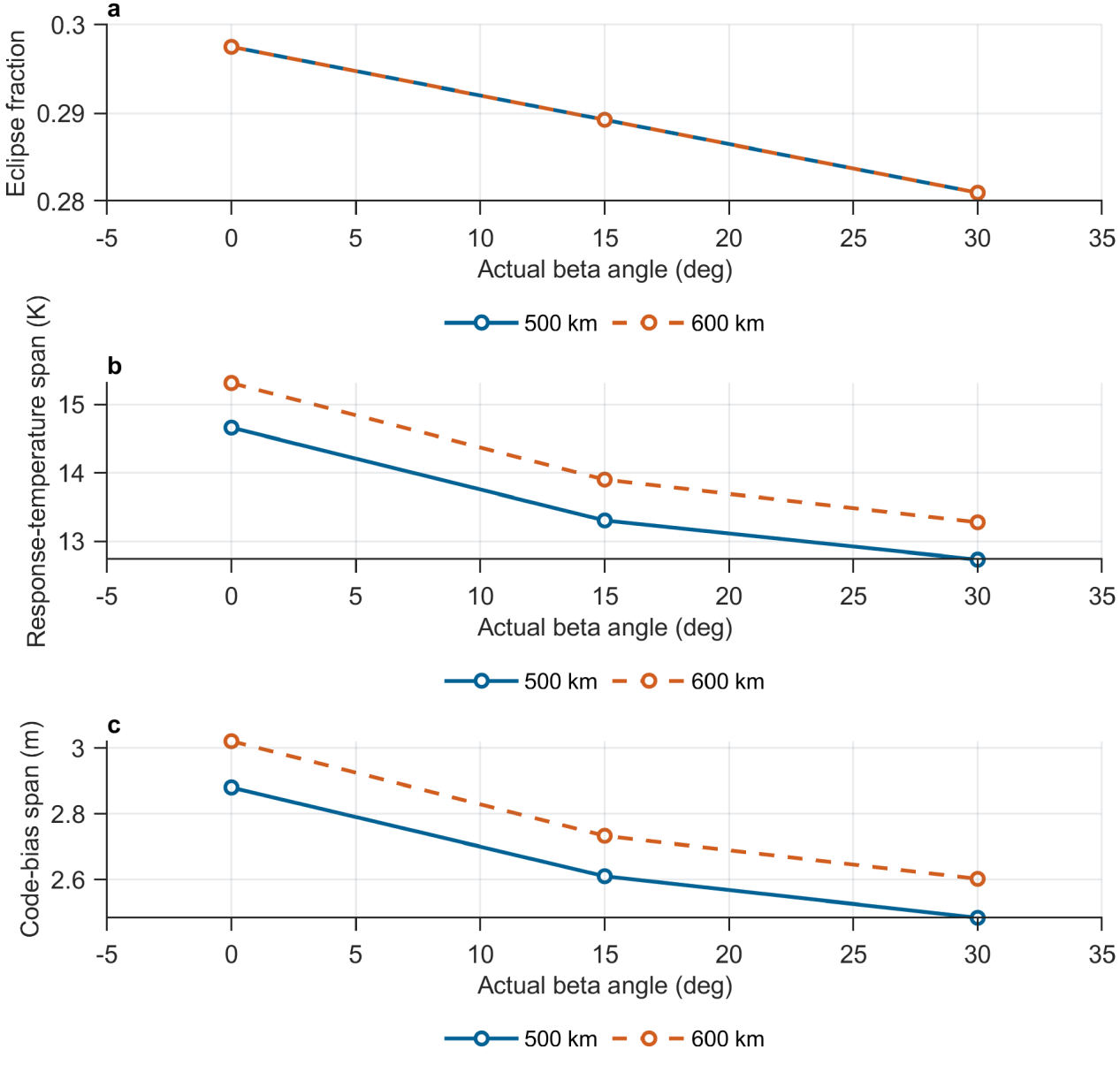
Sweep coverage and response range

6/6 cases completed; beta range -0.0007243 - 30 deg and eclipse fraction 28.10% - 29.75%.

RF temperature-span range: 12.73 - 15.31 K; code-bias-span range: 2.484 - 3.021 m.

Largest code-bias span on this grid: 3.021 m at beta -0.0007243 deg.

Ranges describe this grid; causal effects require controlled comparisons.



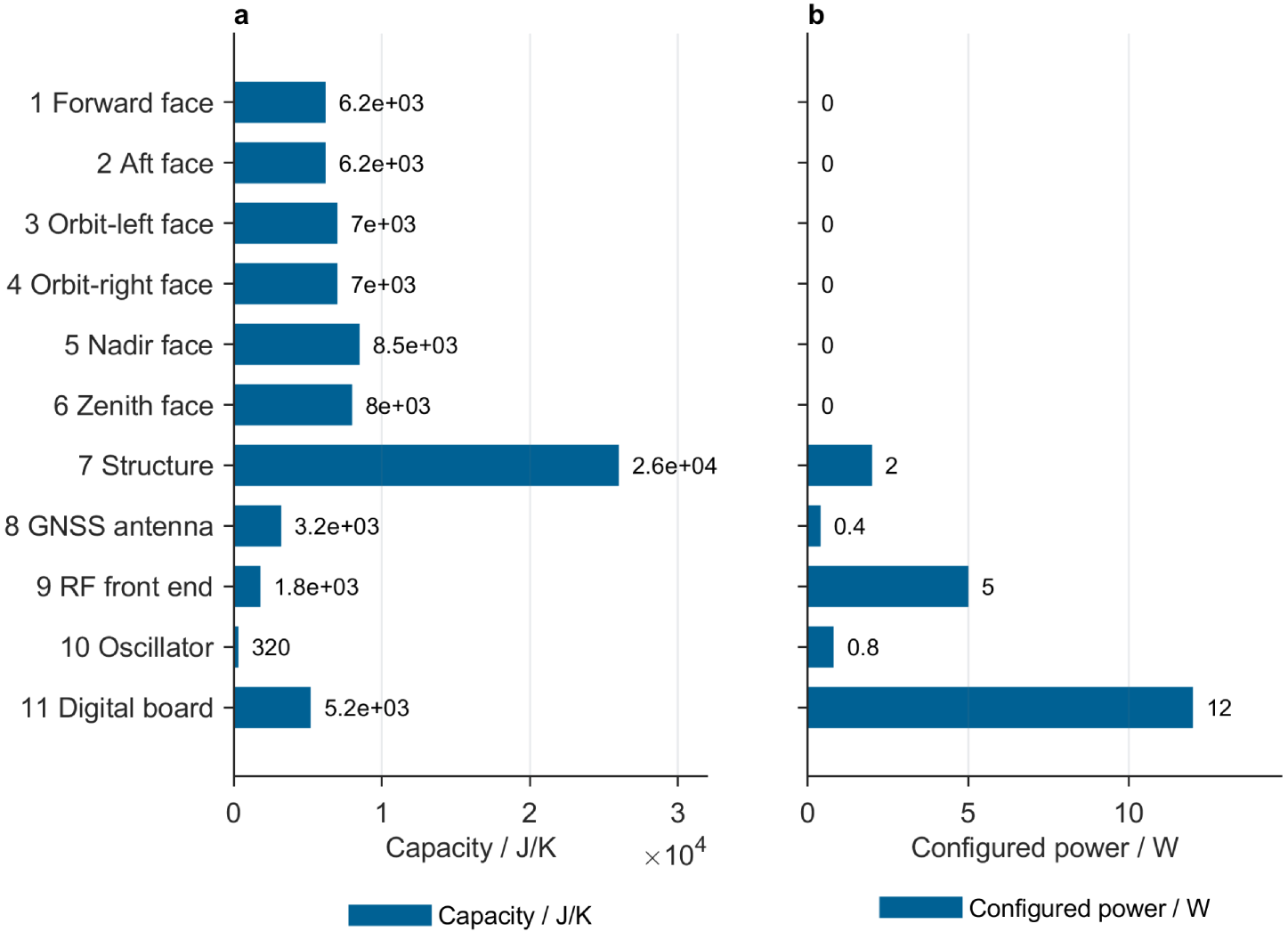
Thermal capacity and internal power

11 nodes; total capacity 79420 J/K; configured internal power 20.2 W.

Spacecraft structure has the largest capacity: 26000 J/K (32.7% of total).

Receiver digital board has the highest configured power: 12 W.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



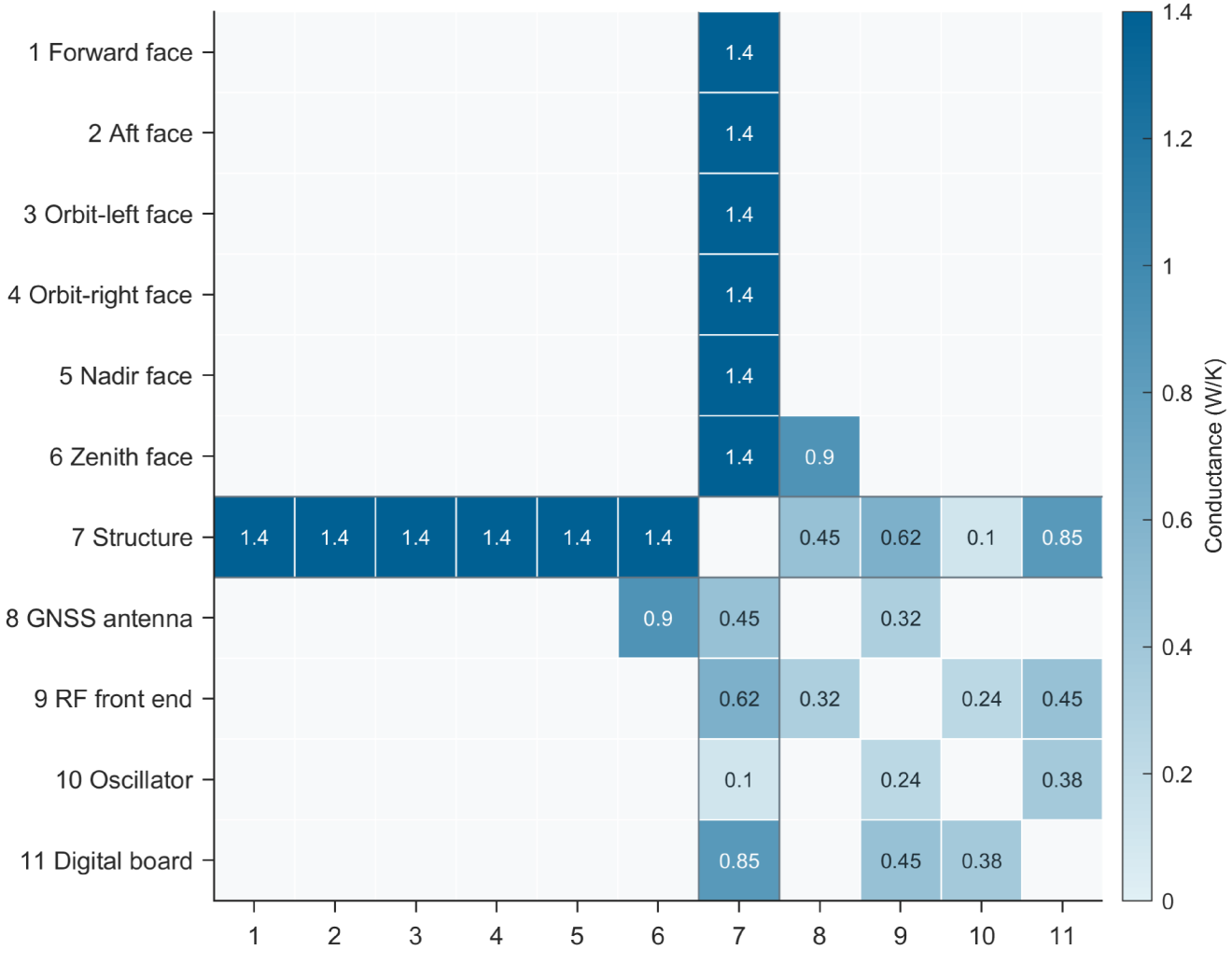
Thermal links and conductance

15 conductive links; Spacecraft structure has the most connections (10).

First 11/11 nodes shown; filled cells and values give conductance. Blank cells have no direct conductive link.

The symmetric matrix shows each reciprocal link twice. Conductances are listed in the link CSV.

Among links connected to GNSS RF front end, the largest absolute mean net transfer is GNSS RF front end to Spacecraft structure at 7.190 W. Total mean net conduction from the response node is 7.260 W (positive outward). This identifies an active path using heat flow rather than link count.

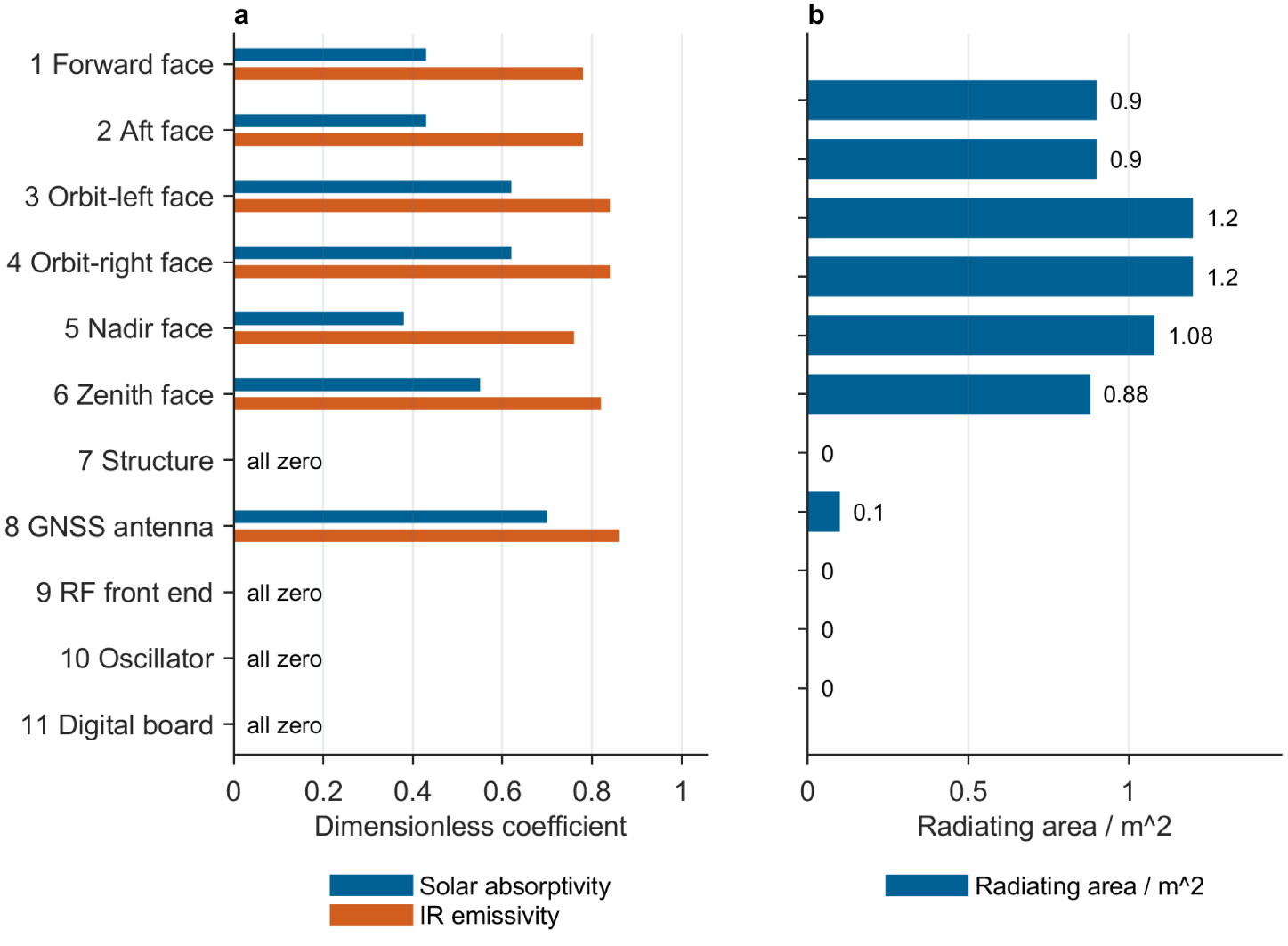


Optical properties and radiating area

Total external radiating area: 6.26 m^2. Zero values are explicitly labelled.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Spacecraft structure, GNSS RF front end, GNSS oscillator, Receiver digital board have zero projected and external radiating areas and exchange heat through conductive links.



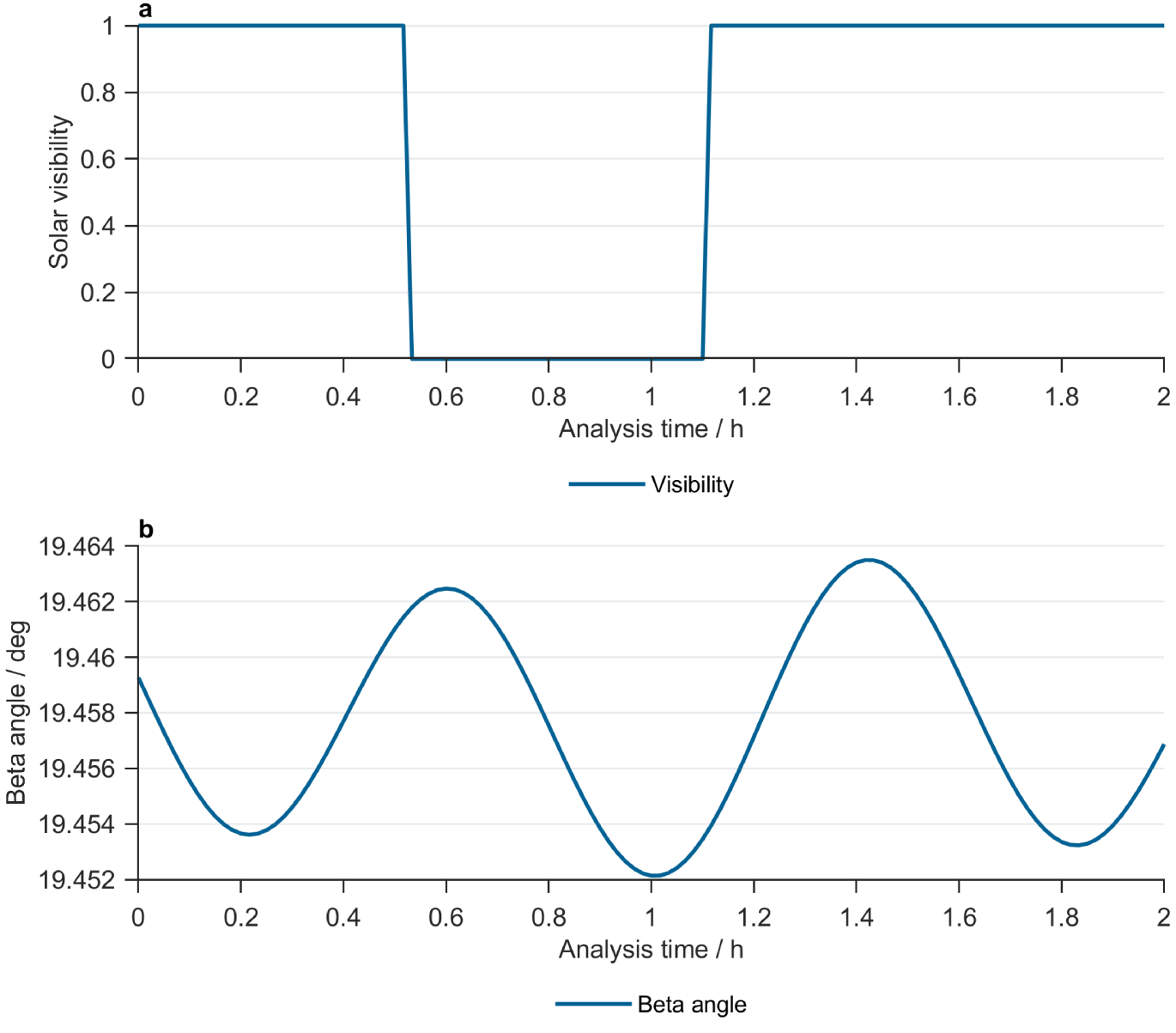
Illumination and eclipse structure

Trapezoidal time weighting estimates partial/full eclipse over 29.17% of the window.

Solar visibility ranges from 0.000 to 1.000; orbital period 96.69 min.

Eclipse uses visibility below 0.999. Time-weighted and epoch-count fractions can differ.

1 eclipse entries and 1 exits; beta angle 19.452 to 19.463 deg.



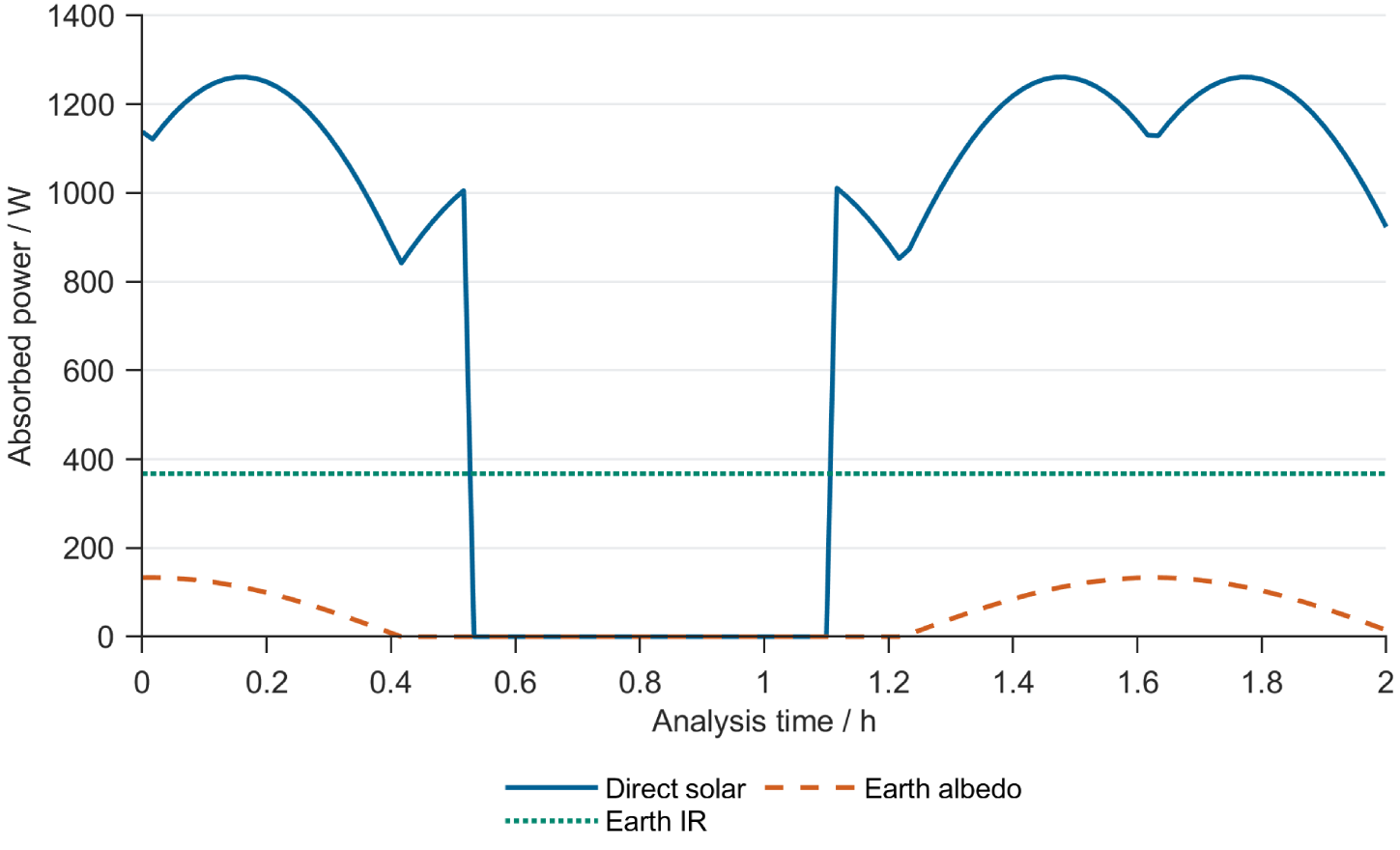
Environmental heat inputs

Time-mean absorbed power: solar 795.415 W; albedo 51.963 W; Earth IR 367.422 W.

Curves sum absorbed power over the network, including area and absorption properties.

Solar absorbed power 0.000 to 1261.402 W; Earth IR 367.422 to 367.422 W.

Time-integrated external heat shares are 65.5% direct solar, 4.3% albedo and 30.2% Earth infrared. direct solar contributes most to external absorption; these are heat-input shares, not temperature-response attribution.



Nodal temperatures and extrema

Window maximum 315.304 K (42.15 deg C, node 6); minimum 220.361 K (-52.79 deg C, node 3).

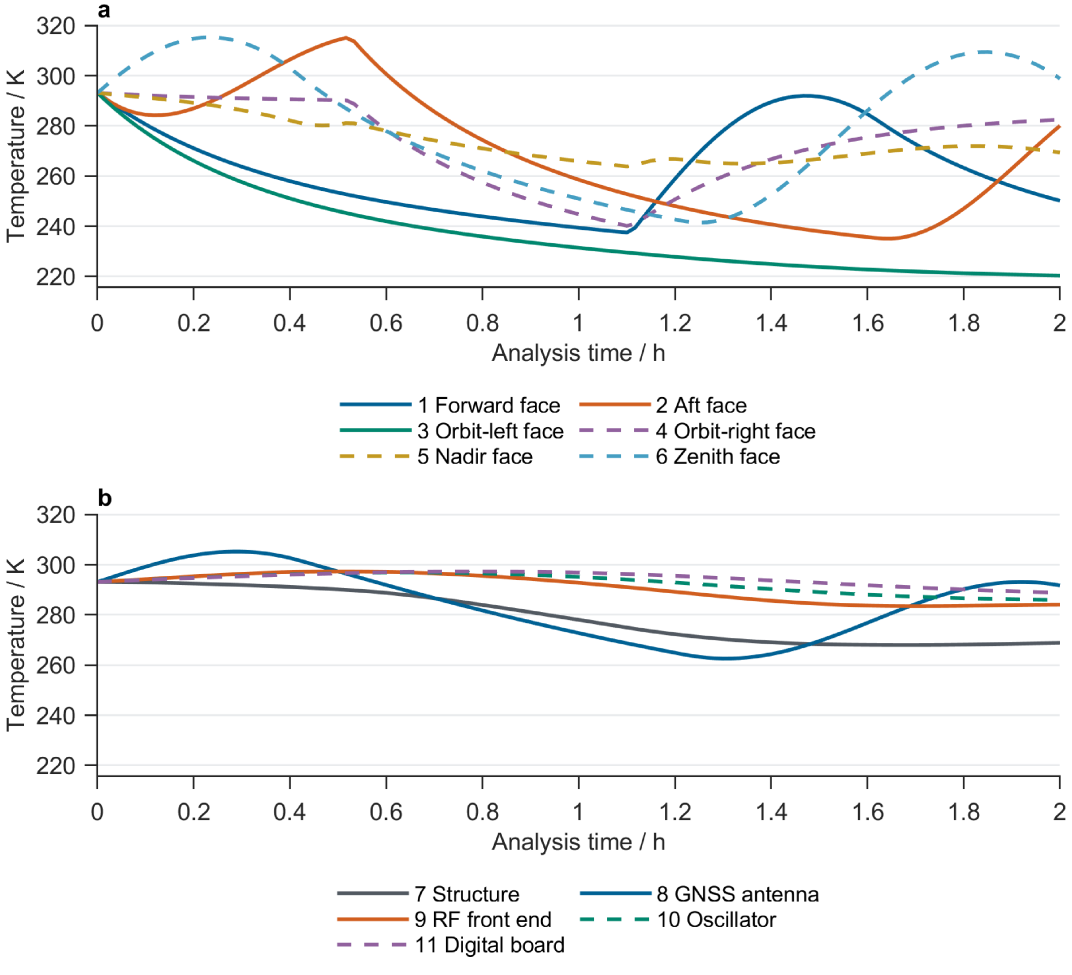
Largest full peak-to-peak range 80.070 K (node 2), distinct from a 95th-5th percentile span.

Hottest node: Zenith face; coldest node: Orbit-left face.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Temperature level and variation describe different demands: the maximum is at Zenith face (315.30 K), the minimum at Orbit-left face (220.36 K), and the largest full range at Aft face (80.07 K).

GNSS RF front end has a full range of 13.805 K, 17.2% of the largest nodal range. Assess response-node stability directly rather than substituting exterior or network-wide extremes.

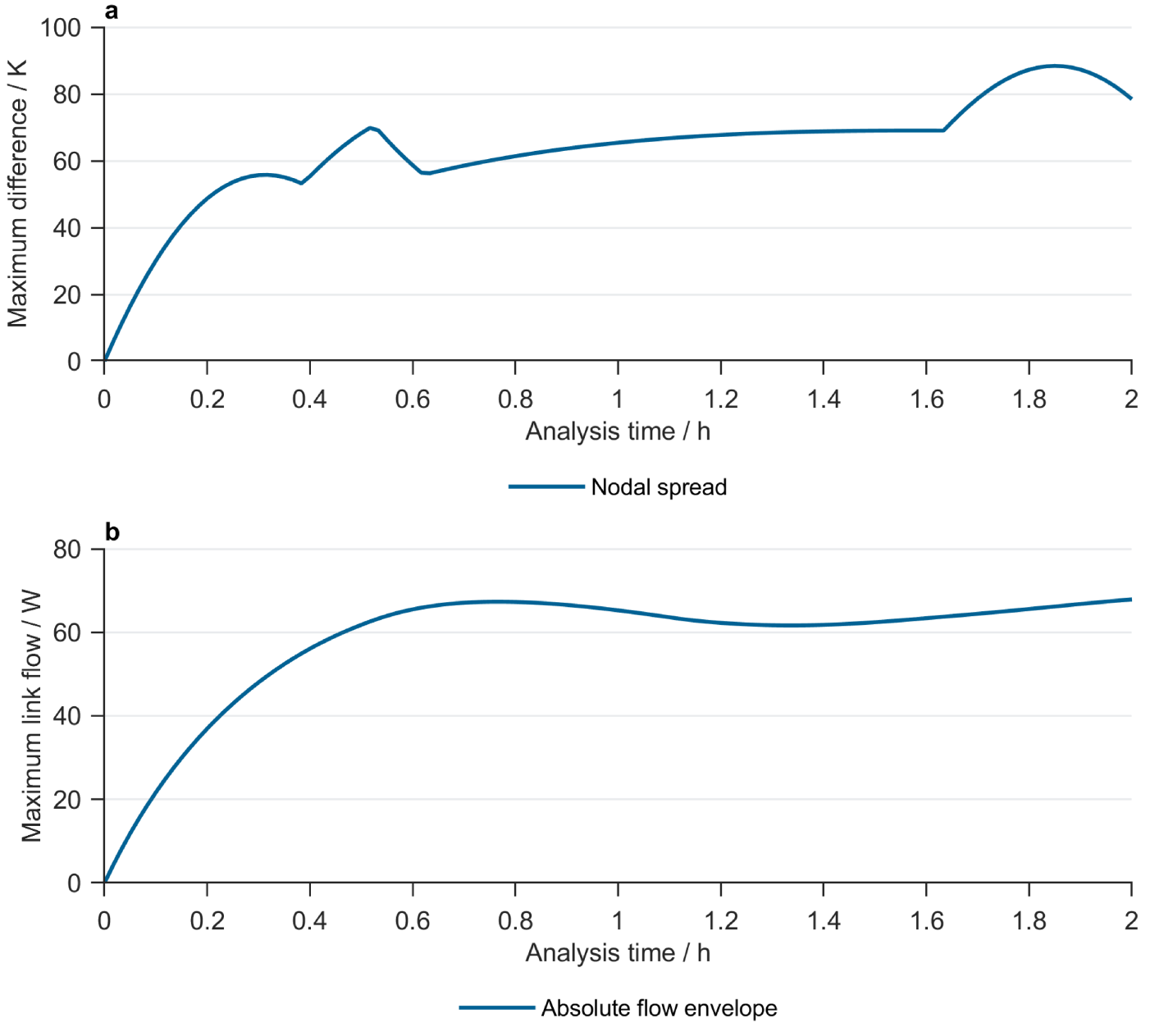


Temperature differences and link heat flow

Maximum simultaneous nodal spread 88.414 K; maximum absolute single-link flow 67.888 W.

Panels show epoch-wise temperature spread and the largest absolute link heat flow.

The envelope takes the largest value across 15 conductive links at each epoch.



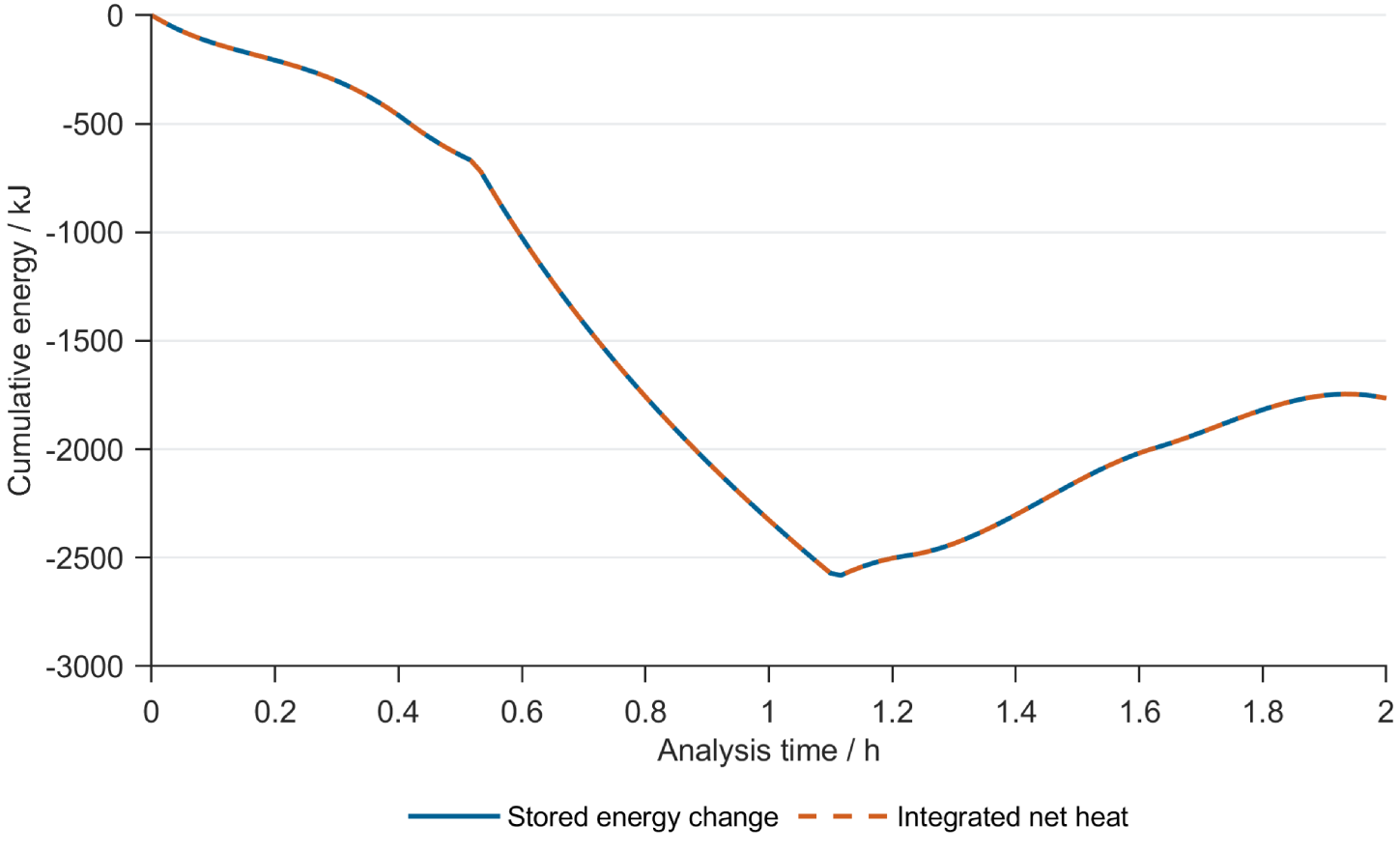
Stored energy and energy balance

After rebasing both at the analysis start, maximum cumulative closure is 6.0536e-09 J.

Stored energy is the sum of capacity times temperature change, relative to analysis start.

Balance: stored energy = external heat + internal heat - radiation + boundary heat.

End-of-window stored-energy change: -1765.6 kJ.



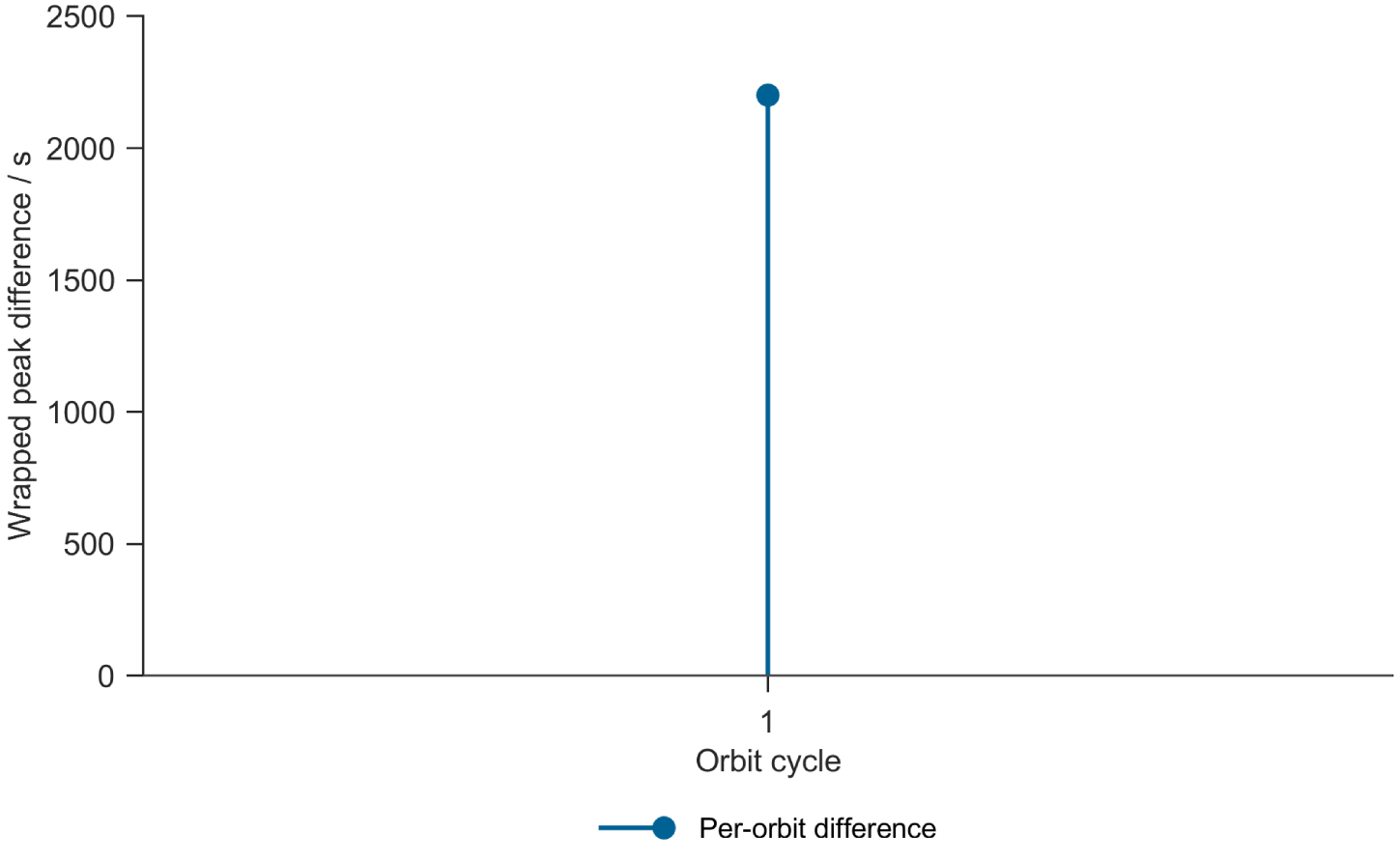
Per-orbit peak timing

Response node: GNSS RF front end.

Peak delay 2201.2 s (1 eligible cycles).

Network external-heat peaks versus response-temperature peaks; differences wrap to +/- half an orbit (2900.6 s).

Positive values indicate later wrapped temperature peaks; negative values indicate earlier peaks. Both are retained.



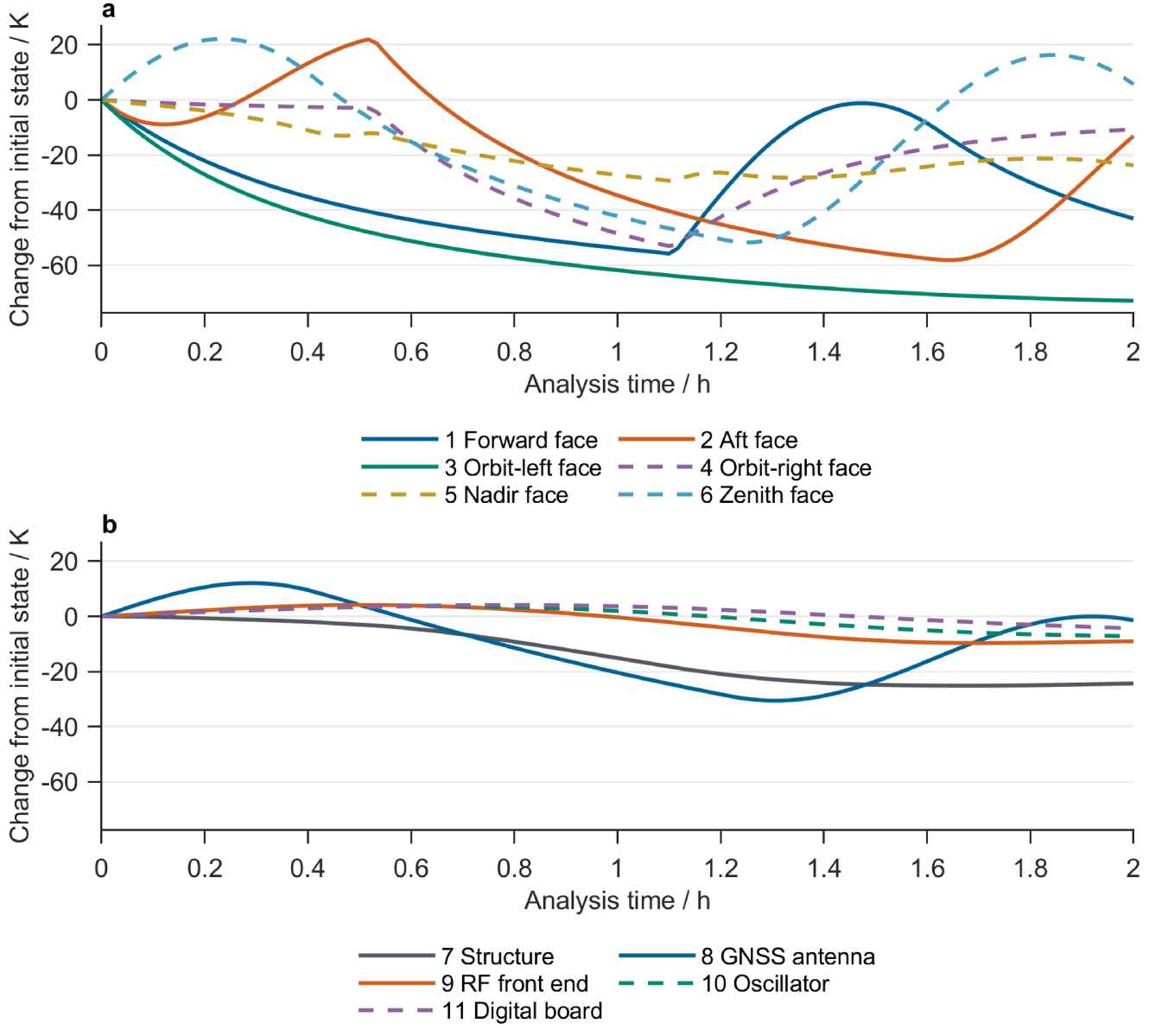
Initial state and temperature drift

Fixed warmup: 0 orbits. Periodic-state convergence was not tested.

Largest absolute end-minus-start temperature change 72.789 K.

Endpoints are 1.241 orbits apart; curves are relative to analysis-start temperatures.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



All-node statistics

Temperature: K; capacity: J/K; configured power: W. Means are time-weighted.

1: Forward face

Min 237.465 | Mean 262.605 | Max 293.150 | Full range 55.685

Capacity 6200 | Internal power 0 | End minus start -42.980 K

2: Aft face

Min 235.058 | Mean 268.154 | Max 315.128 | Full range 80.070

Capacity 6200 | Internal power 0 | End minus start -13.050 K

3: Orbit-left face

Min 220.361 | Mean 237.700 | Max 293.150 | Full range 72.789

Capacity 7000 | Internal power 0 | End minus start -72.789 K

4: Orbit-right face

Min 240.166 | Mean 272.306 | Max 293.150 | Full range 52.984

Capacity 7000 | Internal power 0 | End minus start -10.680 K

5: Nadir face

Min 263.868 | Mean 273.995 | Max 293.150 | Full range 29.282

Capacity 8500 | Internal power 0 | End minus start -23.743 K

6: Zenith face

Min 241.432 | Mean 279.932 | Max 315.304 | Full range 73.872

Capacity 8000 | Internal power 0 | End minus start 5.658 K

All-node statistics (continued)

7: Spacecraft structure

Min 267.987 | Mean 279.282 | Max 293.150 | Full range 25.163

Capacity 26000 | Internal power 2 | End minus start -24.297 K

8: GNSS antenna

Min 262.602 | Mean 284.306 | Max 305.227 | Full range 42.626

Capacity 3200 | Internal power 0.4 | End minus start -1.424 K

9: GNSS RF front end

Min 283.497 | Mean 290.880 | Max 297.302 | Full range 13.805

Capacity 1800 | Internal power 5 | End minus start -9.039 K

10: GNSS oscillator

Min 285.918 | Mean 292.700 | Max 296.859 | Full range 10.941

Capacity 320 | Internal power 0.8 | End minus start -7.232 K

11: Receiver digital board

Min 288.816 | Mean 294.430 | Max 297.306 | Full range 8.490

Capacity 5200 | Internal power 12 | End minus start -4.334 K

Inputs and result details

Scenario name: reference_network_report_demo

Duration: 7200 s

Time step: 60 s

Fixed warmup setting (non-convergence mode): 0 orbit

Node count: 11 node

Conductance range: [0, 1.4] W/K

Internal power range: [0, 12] W

Start epoch UTC: 2024-03-20 12:00:00

Altitude: 600000 m

Inclination: 97.6 deg

Attitude mode: nadir

Trajectory coordinate system: Geocentric equatorial axes (built-in approximate orbit model)

Telemetry comparison: Not run (no telemetry data imported for this task)

Computation checks and audit record

Numerical and configuration checks

Finite scenario temperatures: Pass

Task stages

snapshot: complete

calibration: skipped

freeze model: complete

nodal simulation: complete

batch simulation: complete

geometry simulation: skipped

telemetry comparison: skipped

numerical acceptance: complete

Telemetry comparison: Not run (no telemetry data imported for this task)

Assessment scope

This report evaluates the configured thermal simulation and numerical checks. Device temperature-limit compliance is outside this assessment.

Audit files

MAT results, metric CSV, Markdown, manifest, and figures accompany this PDF.

Software version: 1.2.8

Task conclusions

1. Prioritize the high-temperature level at Zenith face, low-temperature level at Orbit-left face, and variation at Aft face in design review: extrema 315.30/220.36 K and largest range 80.07 K.

2. Prioritize direct solar in environmental-input review: it supplies 65.5% of absorbed external heat. Interpret temperature response together with the heat-transfer paths and orbital phase.

3. The response component GNSS RF front end spans 283.497 to 297.302 K, with a full range of 13.805 K. Use these component values separately from spacecraft extrema.

4. Prioritize the GNSS RF front end / Spacecraft structure contact in conductance review: mean net transfer is 7.190 W. This ranking uses observed model heat flow, not a conductance-perturbation experiment.

5. Sweep coverage and response range: 6/6 cases completed; beta range -0.0007243 - 30 deg and eclipse fraction 28.10% - 29.75%. RF temperature-span range: 12.73 - 15.31 K; code-bias-span range: 2.484 - 3.021 m. Largest code-bias span on this grid: 3.021 m at beta -0.0007243 deg.